

Design & Analysis of a Thick-Walled Pressure Cylinder

Tools Required: Excel (mandatory), calculator

A thick-walled cylinder is subjected to internal pressure. The cylinder has the following dimensions and material properties:

- Internal radius: $a = 50$ mm
- External radius: $b = 100$ mm
- Internal pressure: $p_i = 30$ MPa
- External pressure: $p_o = 5$ MPa

Group	Material
1	Carbon Steel
2	Aluminium alloy 6061-T6
3	Titanium alloy Ti-6Al-4V
4	Brass

Question

- Calculate hoop stresses at intervals of 5 mm across the cylinder wall
- Calculate Pressure distribution at intervals of 5 mm across the cylinder wall
- Plot the following:
 - Hoop stress vs radius
 - Pressure vs radius

Highlight the **maximum hoop stress** on the plot.
Add proper **axis titles, legend, and chart title**.

Submission Requirements

- PDF or Word report with derivations and analysis
- Excel file with table and plots
- Hand-calculated stress at **minimum, maximum, and midpoint** of the cylinder wall
- Short discussion (1 paragraph) on stress variation